

# Python and Table Readiness

Project: Responsible Plant Classifier

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Classification or Regression?

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### The ML Lifecycle

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Classify records for Rows columns and schemas. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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### 3. Individual reasoning check

Use the worked case for Rows columns and schemas. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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### 4. Independent transfer case

**Independent case: Changed independent case**

Supplied values are 3, 7, 14. Compute their mean. What should a function report for an empty list? How should it handle a missing fourth record?

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

Trace mean for [4,8,12] using `sum(values)/len(values)`. What should the program do for []? Explain why returning 0 silently is misleading.

## 6. Solve the given case

Python `lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths)`.

Trace and handle an empty list.

Show your trace, calculation or evidence

**Write or draw your reasoning and evidence.**

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## 7. Check a changed case

Trace Python: `lengths = [4, 7, 9]; selected = lengths[1]; total = selected + 3`. Give selected and total.

My independent answer and reason

**Write or draw your reasoning and evidence.**

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## Week 4 Test and reflect

### 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

#### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

### 9. Project checkpoint

Create a tiny five-row plant table with explicit column names, types, units, and a reproducible summary calculation.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

Trace mean for [4,8,12] using `sum(values)/len(values)`. What should the program do for []? Explain why returning 0 silently is misleading.

### My independently reasoned answer

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Record support used: none, hint or teacher explanation.

## 12. W1 State a schema

List the required fields, types and duration boundaries for the worked table.

### My reasoning and evidence

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## 13. W2 Classify records

Classify minutes values 0, 60, -1 and 61 under this schema.

### My reasoning and evidence

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## 14. W3 Repair a unit problem

A source says 180 seconds but the row says minutes: 180. Give a documented correction.

### My reasoning and evidence

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## 15. W4 Design a schema independently

For a fictional shelf audit, define label and book\_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

### My reasoning and evidence

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## 16. W5 Separate validity and truth

A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?

### My reasoning and evidence

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## 17. File input and output — W1 Trace types

Before and after `int("8")`, identify the types and values.

### My reasoning and evidence

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## 18. File input and output — W2 Predict the total

A sandbox file has lines 5, 0 and 2. What total does the worked reader print?

### My reasoning and evidence

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## 19. File input and output — W3 Repair a path issue

The file is in the class-data folder but the script looks in its working folder. Give a safe repair.

### My reasoning and evidence

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## 20. File input and output — W4 Read independently

In a fresh sandbox file counters.txt put 4, 6 and 0 on separate lines. Adapt the reader, print the number of values and total. State the expected results.

### My reasoning and evidence

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## 21. File input and output — W5 Handle malformed data

The file contains twelve on one line. Why should the program not silently turn it into zero?

**My reasoning and evidence**

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## 22. Exceptions and recovery — W1 Trace success

What does `parse_minutes("0")` return, and why is zero allowed?

**My reasoning and evidence**

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## 23. Exceptions and recovery — W2 Trace two failures

Compare `parse_minutes("abc")` and `parse_minutes("70")`.

**My reasoning and evidence**

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## 24. Exceptions and recovery — W3 Repair a handler

An except block returns 0 for any failure. State two problems and a repair.

### My reasoning and evidence

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## 25. Exceptions and recovery — W4 Build independently

Adapt the function for count strings representing integers 0 through 10. Test "4", "0", "11", "four" and "-1" and record status/value.

### My reasoning and evidence

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## 26. Exceptions and recovery — W5 Choose recovery

A batch has 8 valid and 2 invalid lines. What should its report include?

### My reasoning and evidence

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## 27. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 28. Pictorial case practice: Python and Table Readiness

Inspect a small table before asking Python to compute an average.

Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/4

## Python and Table Readiness

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Check records

Four rows contain three measured values and one missing value.



#### 2. Calculate mean

Measured sum is  $2+6+0=8$ ; mean is  $8/3$  cm, approximately 2.67 cm.



#### 3. Report coverage

Report 3/4 measured, retaining the missing record separately.

### CHANGE ONE THING

S changes from 0 to None.

Explain the changed result using the matching lesson.

Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.